Staggered kite finite elements for the shallow-water equations and the compressible Euler equations on polygonal meshes

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Abstract

Every polygonal mesh of the sphere or the plane — triangles, quadrilaterals and the hexagons and pentagons of Voronoi meshes — is split into quadrilaterals (*kites*) by joining the cell centres with the edge points. On the kite grid we define a height space of tensor Gauss–Radau elements, continuous inside a primal cell and broken across primal edges, and a velocity space of Raviart–Thomas elements, continuous in the normal component inside a dual cell and broken across its boundary. The height mass matrix is diagonal, the velocity mass matrix block diagonal with one block per dual cell, and the divergence and gradient are exactly adjoint and metric free. The resulting shallow-water scheme conserves mass and energy exactly and is second-order accurate on all mesh types. The extension to the nonhydrostatic compressible Euler equations is a direct product with the layers of a height-based vertical grid: one implicit column per height node and an explicit horizontal step (HEVI). Its horizontal transport needs care. We identify three instabilities, analyse them with eigenvalues of the linearized tendency, and remove them: a quadrature mismatch between the velocity mass and the projections, a projection defect of the transport of the potential temperature, and an instability of the vector-invariant momentum transport when the wind is faster than the internal gravity waves, which in a stratified atmosphere it always is for the higher vertical modes. A discontinuous Galerkin transport of momentum, density and potential temperature with the pointwise velocity removes it. The model then reproduces the linear growth rate of baroclinic instability in an f-plane channel to within a few per cent and gives the same baroclinic wave on hexagonal, cubed-sphere and triangular meshes. Everything is implemented in the Julia package `dmplex.jl`.

Contents

1 Introduction **2**

2 The kite grid **2**

3 Finite-element spaces **3**

4 Discrete operators **4**

5 Shallow-water schemes **4**

6 Shallow-water results **5**

7 The 3D compressible model **6**

7.1 Equations 6

7.2 Spaces 7

7.3 Horizontal terms: the vector-invariant form 7

7.4 Vertical terms and time stepping 8

8 Stability of the horizontal transport **8**
8.1 Quadrature of the velocity mass 8
8.2 Projected transport of θ 8
8.3 Supercritical flow 9

9 3D results **10**
9.1 Thermal wind 10
9.2 Baroclinic wave in an f-plane channel 11
9.3 Baroclinic wave on the sphere 11

10 Conclusions **12**

1 Introduction

Compatible (mimetic) finite elements give the discrete operators of geophysical fluid dynamics the structure of the continuous ones: the gradient is the negative adjoint of the divergence, the curl of a gradient vanishes, and the semi-discrete equations conserve mass and energy [1, 3, 4]. The price is that the mass matrices of the usual pairs (e.g. $\mathrm{RT}_k \times \mathrm{DG}_k$) are not diagonal, so every explicit step needs global solves. Finite-volume C grids such as TRiSK [6, 5] and ICON [12] avoid them with diagonal Hodge stars, at the price of first-order operators on general meshes.

The dual cell method of Wess, Kapidani, Codecasa and Schöberl [9] splits every primal cell into quadrilaterals around the primal vertices and uses spaces that are continuous inside primal or dual cells only. The mass matrices become diagonal or block diagonal, and the order can be raised. This paper transfers that idea to the rotating shallow-water equations on arbitrary polygonal meshes of the sphere (Sections 2–6) and then to the nonhydrostatic compressible Euler equations (Sections 7–9). The 2D model conserves mass and energy exactly. The 3D model is a direct product with the layers of a vertical grid, so it keeps the column structure that a horizontally explicit, vertically implicit (HEVI) time step needs.

The 3D extension is not straightforward. Section 8 records three horizontal instabilities that the 2D tests do not show, their analysis with eigenvalues of the linearized tendency and their removal. The last one is the most important: the energy-conserving vector-invariant transport is unstable when the flow is faster than the gravity waves, and in a stratified atmosphere the internal waves of the higher vertical modes are always slower than the wind. A discontinuous Galerkin (DG) transport with the pointwise velocity removes it. With it the model is stable on the steady thermal wind, reproduces the linear baroclinic growth rate in an f-plane channel and gives the same baroclinic wave on three different meshes.

2 The kite grid

Let $\mathcal{P}$ be a polygonal primal mesh of a closed surface Ω (the unit sphere or a doubly periodic plane). Every cell $P \in \mathcal{P}$ with vertices $w_1, \dots, w_n$ is split into n kites

$$K_i = (w_i,\ m_i,\ c_P,\ m_{i-1}), \qquad i = 1, \dots, n,$$

where c_P is the cell centre and m_i a point on the primal edge (w_i, w_{i+1}) (Figure 1). The kites around a primal vertex w form its dual cell $\widetilde{P}_w$. Every kite has the same local numbering: vertex 1 is the primal vertex and vertex 3 the centre. Local edges 1 and 4 are halves of primal edges (*primal faces*). Local edges 2 and 3 join an edge point to the centre (*dual faces*); they form the boundaries of the dual cells.

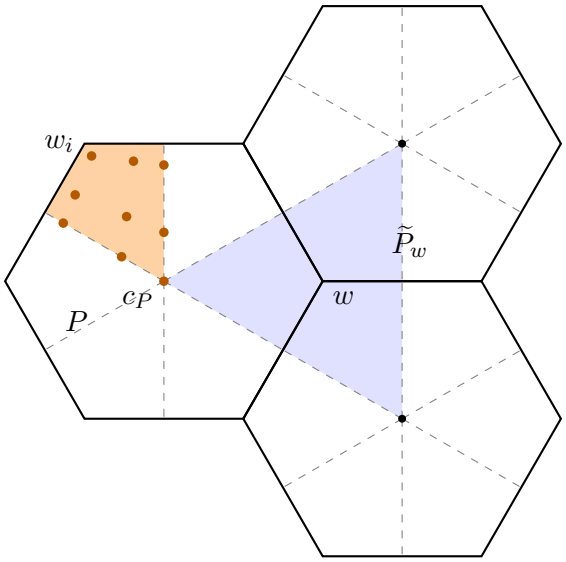


Figure 1: Kite grid of a hexagonal mesh: primal edges (solid), dual faces from the centres to the edge points (dashed). Shaded blue: the dual cell $\widetilde{P}_w$ of the vertex w, made of the three kites that touch it. Orange: one kite of the cell P with the nine Gauss–Radau nodes of the Q_2 height space; they lie inside the kite, on its dual faces and at the centre, never on primal edges.

Cell centres. The centre is the vertex of the orthogonal dual mesh when there is one: the circumcentre of a triangle, the generator of a Voronoi cell. The generators of a Voronoi mesh are recovered from the cells as the null vector of the homogeneous linear system $g_i - g_j \parallel \mathbf{n}_e$, $\mathbf{n}_e \cdot (g_i + g_j) = 0$ over all edges e between cells i, j; for MPAS and ICON meshes this reproduces the stored generators to 10^{-14}. On other meshes (cubed sphere, planes) the centre is the centroid. With an orthogonal dual, m_i is the crossing of the primal edge with the dual edge, so the dual faces lie on the dual edges; otherwise m_i is the edge midpoint.

3 Finite-element spaces

All kite spaces live on the reference square $[-1, 1]^2$ with the primal vertex at $(-1, -1)$ and the centre at $(1, 1)$. The reference map is the bilinear interpolant of the four kite vertices, on the sphere radially projected (curved elements).

Height. $H_p = \texttt{KiteRadau}(p)$: Q_p on every kite with the nodes at the tensor Gauss–Radau points $r_0 < \cdots < r_p = 1$ that include the end point at the centre side. The nodes lie inside the kite (p^2), on its dual faces (p each) and at the centre, never on primal edges. So a height field is continuous inside a primal cell and broken across primal edges,

$$H_p = \{h \in L^2(\Omega) : h|_P \in H^1(P)\ \forall P \in \mathcal{P},\ h|_K \in Q_p\},$$

with $1 + n_P(p + p^2)$ degrees of freedom per primal cell (37 for $p = 2$ on a hexagon). With the Radau nodes as quadrature the mass matrix M_h is diagonal; for $p \geq 1$ the rule integrates the bilinear area element exactly, so the lumped mass is consistent.

Velocity. $U_k = \texttt{KiteBrokenRT}(k)$: RTCF_k on every kite, mapped with the contravariant Piola transform $\mathbf{v} = J\hat{\mathbf{v}}/\det J$, with normal continuity across the primal faces (inside a dual cell) and none across the dual faces:

$$U_k = \{\mathbf{u} \in L^2(\Omega)^2 : \mathbf{u}|_{\widetilde{P}} \in H(\mathrm{div}, \widetilde{P})\ \forall \widetilde{P} \in \widetilde{\mathcal{P}},\ \mathbf{u}|_K \in \mathrm{RTCF}_k\}.$$

The degrees of freedom of the dual faces belong to the kite itself. The mass matrix M_u is block diagonal with one block per dual cell (30 for RT_1 on the triangular dual cell of a hexagonal mesh). Two alternatives with the same continuity pattern are implemented: broken BDM_k, and the $Q_p \times Q_p$ space of Wess et al. [9] with Radau nodes at the primal-vertex side, whose lumped mass couples only the components at one node.

Vorticity. Z_{k+1}: continuous Q_{k+1} on the kites with Gauss–Lobatto nodes and lumped mass. The default pair is $\mathrm{RT}_1 \times Q_2 \times \mathrm{CG}_2$ ($k = 1$, $p = 2$).

4 Discrete operators

With the contravariant Piola map one has $\mathbf{v} \cdot \nabla\eta \, dA = \hat{\mathbf{v}} \cdot \hat{\nabla}\hat{\eta} \, d\xi$, $(\nabla \cdot \mathbf{v})\, \eta \, dA = (\hat{\nabla} \cdot \hat{\mathbf{v}})\, \hat{\eta} \, d\xi$ and $\mathbf{v} \cdot \mathbf{n} \, ds = \hat{\mathbf{v}} \cdot \hat{\nu} \, dt$, so both operators below are free of the metric.

Divergence. For $\eta \in H_p$ and $\mathbf{F} \in U_k$

$$D(\eta; \mathbf{F}) = \sum_K \int_K \mathbf{F} \cdot \nabla\eta - \sum_K \sum_{e \subset \partial K \cap \partial\mathcal{P}} \int_e \eta_K \, \mathbf{F} \cdot \mathbf{n}_K, \tag{1}$$

summed over the primal faces of each kite with the trace of η from inside K. As a matrix, $B_{ij} = D(\eta_i; \mathbf{v}_j)$.

Gradient. For $p \in H_p$ and $\mathbf{v} \in U_k$

$$G(p; \mathbf{v}) = \sum_K \int_K p \, \nabla \cdot \mathbf{v} - \sum_K \sum_{e \subset \partial K, \; e \text{ dual}} \int_e p \, \mathbf{v}_K \cdot \mathbf{n}_K,$$

where p is single valued on the dual faces.

Proposition 1. *$G(p; \mathbf{v}) = -D(p; \mathbf{v})$, i.e. $G = -B^T$, and $\mathbf{1}^T B = 0$.*

Proof. On each kite $\int_K p \, \nabla \cdot \mathbf{v} = -\int_K \nabla p \cdot \mathbf{v} + \int_{\partial K} p \, \mathbf{v} \cdot \mathbf{n}_K$. The boundary integral over the dual faces cancels the face term of G, leaving $-\sum_K \int_K \nabla p \cdot \mathbf{v} + \sum_K \sum_{\text{primal}} \int p_K \, \mathbf{v} \cdot \mathbf{n}_K = -D(p; \mathbf{v})$. For $\eta \equiv 1$, $D(1; \mathbf{v}) = -\sum_K \sum_{\text{primal}} \int \mathbf{v} \cdot \mathbf{n}_K = 0$, since $\mathbf{v} \cdot \mathbf{n}$ is continuous across primal faces and both sides appear with opposite normals. □

Vorticity, kinetic energy, Coriolis. $\langle \zeta, \gamma \rangle = -\int \mathbf{u} \cdot (\hat{\mathbf{r}} \times \nabla\gamma)$ for $\gamma \in Z_{k+1}$, $\langle K, \eta \rangle = \int \frac{1}{2}|\mathbf{u}|^2 \eta$ in H_p, and the antisymmetric $C_{ij} = \int f \, \mathbf{v}_i \cdot (\hat{\mathbf{r}} \times \mathbf{v}_j)$, with $\hat{\mathbf{r}} = J_1 \times J_2 / |J_1 \times J_2|$.

5 Shallow-water schemes

The rotating shallow-water equations in vector-invariant form are

$$\partial_t \mathbf{u} + q \, \hat{\mathbf{r}} \times \mathbf{F} + \nabla(K + gh) = 0, \qquad \partial_t h + \nabla \cdot \mathbf{F} = 0, \tag{2}$$

with the mass flux $\mathbf{F} = h\mathbf{u}$, $K = \frac{1}{2}|\mathbf{u}|^2$ and the potential vorticity $q = (\zeta + f)/h$. They conserve mass and the energy $E = \int \frac{1}{2} h |\mathbf{u}|^2 + \frac{1}{2} g h^2$.

Linear. $M_h \dot{h} = H \, B \, \mathbf{u}$, $M_u \dot{\mathbf{u}} = -C\mathbf{u} - g \, B^T h$ conserves mass ($\mathbf{1}^T B = 0$) and the energy $\frac{1}{2} H \mathbf{u}^T M_u \mathbf{u} + \frac{1}{2} g h^T M_h h$ (C antisymmetric, $G = -B^T$).

Nonlinear. The mass flux is projected into the velocity space, $\mathbf{F} = \Pi_U(h\mathbf{u})$, i.e. $M_u\mathbf{F} = (\int h\mathbf{u}\cdot\mathbf{v}_i)_i$, one solve with the block-diagonal M_u, and q is evaluated at the quadrature points:

$$M_h\dot{h} = B\mathbf{F}, \qquad \int \dot{\mathbf{u}}\cdot\mathbf{v} = -\int q\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{v} - B^T(gh+K). \tag{3}$$

Proposition 2. *The scheme* (3) *conserves mass and the discrete energy* $E_h = \int \frac{1}{2}h|\mathbf{u}|^2 + \frac{1}{2}g\,h^T M_h h$ *exactly.*

Proof. Mass: $\mathbf{1}^T M_h\dot{h} = \mathbf{1}^T B\mathbf{F} = 0$. Energy: with the same quadrature in E_h and the right-hand sides, $\partial E_h/\partial\mathbf{u} = M_u\mathbf{F}$ and $\partial E_h/\partial h = M_h(K+gh)$, so $\dot{E}_h = -\int q\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{F} - \mathbf{F}^T B^T(gh+K) + (gh+K)^T B\mathbf{F} = 0$. □

This is the framework of McRae and Cotter [3] on the kite spaces, with mass matrices that stay diagonal or block diagonal. The nonlinear terms use one Gauss rule per kite (3×3 for RT_1) with precomputed metric terms; energy conservation is an algebraic property and holds for every rule. The terms are KernelAbstractions kernels over the kites with atomic scatter and run unchanged on CPUs and GPUs; the model is distributed with MPI over equal-area partitions of the primal cells, one ring of ghost cells deep.

6 Shallow-water results

Operator consistency. L^2 errors of the discrete gradient $M_u^{-1}B^T\Pi p$ and divergence $M_h^{-1}B\,\Pi\mathbf{u}$ against the projections of the exact operators, for smooth p and $\mathbf{u}$, on hexagon kites (960, 3840, 15 360 kites) and cubed-sphere kites (384, 1536, 6144):

pair	mesh	gradient	divergence
$Q_0\times\mathrm{RT}_0$	hexagons	7.4, 7.6, 7.7	0.017, 0.0069, 0.0030
	cube	5.6, 5.8, 5.9	0.065, 0.076, 0.067
$Q_1\times\mathrm{RT}_0$	hexagons	0.10, 0.052, 0.026	0.23, 0.21, 0.21
	cube	0.19, 0.094, 0.047	0.40, 0.36, 0.31
$Q_1\times\mathrm{RT}_1$	hexagons	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027
	cube	0.31, 0.15, 0.077	0.044, 0.022, 0.012
$Q_2\times\mathrm{RT}_1$	hexagons	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023
	cube	0.0091, 0.0023, 0.00057	0.059, 0.035, 0.019

At the lowest order one of the two operators is inconsistent: with Q_0 heights the pressure is constant per primal cell and does not control the velocity degrees of freedom of the dual faces, and the divergence of broken RT_0 tested with Q_1 does not converge. From RT_1 on both converge.

Williamson test case 2. The steady geostrophic flow [11] converges at second order on all meshes. Linear model, $Q_1\times\mathrm{RT}_1$, RK4, $\|h(T)-h(0)\|$:

primal mesh	coarse	medium	fine
hexagons (162, 642, 2562 cells)	$3.09\cdot10^{-3}$	$7.72\cdot10^{-4}$	$1.94\cdot10^{-4}$
triangles (80, 320, 1280)	$9.67\cdot10^{-3}$	$2.46\cdot10^{-3}$	$6.16\cdot10^{-4}$
cubed sphere (96, 384, 1536)	$7.88\cdot10^{-3}$	$1.94\cdot10^{-3}$	$4.88\cdot10^{-4}$

Nonlinear model, $\mathrm{RT}_1\times Q_2\times\mathrm{CG}_2$, SSP-RK3: $3.60\cdot10^{-4}$ and $8.80\cdot10^{-5}$ on hexagons 2 and 3 (rate 2.03), mass and energy conserved to round-off. Flat (chordal) kites give rate 1.87. The velocity pairs compare as follows (linear geostrophic state with Coriolis, finest hexagon mesh, rate in brackets):

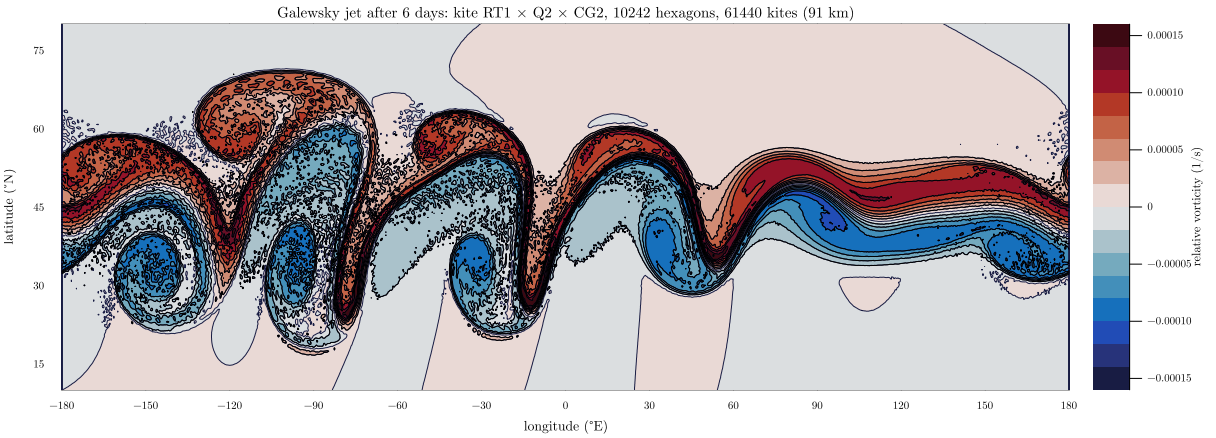


Figure 2: Galewsky jet, relative vorticity after 6 days, kite method $\mathrm{RT}_1 \times Q_2 \times \mathrm{CG}_2$ on 10 242 hexagons (61 440 kites), no dissipation.

pair (velocity / height)	gradient	divergence	geostrophic
RT_1 / Q_1	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027	$1.94 \cdot 10^{-4}$ (1.99)
RT_1 / Q_2	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023	$4.12 \cdot 10^{-5}$ (2.10)
BDM_1 / Q_1	0.13, 0.065, 0.032	0.28, 0.26, 0.25	$1.46 \cdot 10^{-4}$ (1.78)
Wess et al., $p = 1$	0.14, 0.069, 0.035	0.21, 0.18, 0.18	$1.68 \cdot 10^{-4}$ (1.86)
Wess et al., $p = 2$	0.0070, 0.0018, 0.00044	0.026, 0.013, 0.0064	$5.61 \cdot 10^{-6}$ (2.07)

Comparison with the mixed method on the kite quadrilaterals. The kite grid is an ordinary quadrilateral mesh, so the compatible pair $\mathrm{RT}_1 \times \mathrm{DG}_1 \times \mathrm{CG}_2$ with global mass matrices runs on it unchanged. On hexagon kites 2 and 3 it is about 1.9× more accurate ($1.94 \cdot 10^{-4}$, $4.72 \cdot 10^{-5}$) with fewer degrees of freedom, but needs a global sparse factorization; the kite method replaces it by independent dual-cell blocks, which matters most for explicit time stepping, on GPUs and in parallel.

Galewsky jet. The barotropically unstable mid-latitude jet of Galewsky et al. [2], without diffusion, on 10 242 hexagons (61 440 kites, about 91 km), SSP-RK3 with $\Delta t = 33.5$ s (Figure 2). The jet breaks up after day 4 and rolls up into the familiar vortices by day 6. After 6 days the relative change of mass is $1.2 \cdot 10^{-11}$ and of energy $9 \cdot 10^{-10}$, from the time integration only. Without any dissipation, grid-scale noise in the vorticity accompanies the vortices.

Bickley jet. A doubly periodic $[0, 4\pi)^2$, $u = \operatorname{sech}^2(y - 2\pi)$, no rotation, Froude number 0.2, perturbed in wavenumber 1. On 64×64 quadrilaterals (16 384 kites) the jet forms a vortex street with two cat's-eye pairs by $t = 40$ (Figure 3); mass and energy change by $1.6 \cdot 10^{-10}$ and $7 \cdot 10^{-11}$.

7 The 3D compressible model

7.1 Equations

The dry compressible Euler equations of a shallow atmosphere over flat ground, in height coordinates, with the prognostic variables $\mathbf{u}$ (horizontal velocity), w, ρ and $\rho\theta$ and the Exner pressure $\pi = (R\rho\theta/p_0)^{R/c_v}$:

$$\partial_t \mathbf{u} = -(\zeta + f)\,\hat{\mathbf{r}} \times \mathbf{u} - w\,\partial_z \mathbf{u} - \nabla K - c_p \theta \nabla \pi, \qquad \partial_t w = -\mathbf{u} \cdot \nabla w - w\,\partial_z w - c_p \theta\,\partial_z \pi - g, \quad (4)$$

$$\partial_t \rho = -\nabla \cdot (\rho \mathbf{u}) - \partial_z (\rho w), \qquad \partial_t (\rho\theta) = -\nabla \cdot (\rho\theta \mathbf{u}) - \partial_z (\rho\theta w), \quad (5)$$

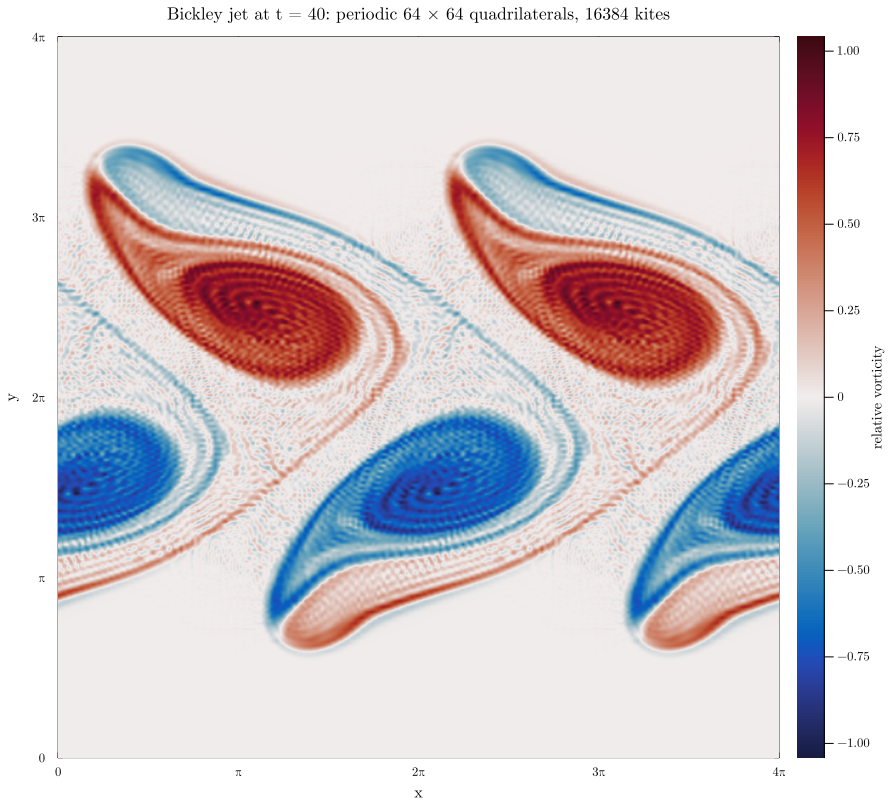


Figure 3: Bickley jet, relative vorticity at $t = 40$ on 64×64 quadrilaterals (16 384 kites), no dissipation.

with $K = \frac{1}{2}|\mathbf{u}|^2$; Rayleigh damping of w acts near the lid.

7.2 Spaces

The spaces are tensor products of the 2D kite spaces with the layers of a vertical grid (thicknesses Δz_k, centres z_k, interfaces $z_{j-1/2}$, $\Delta z^h_j = z_j - z_{j-1}$):

field	space	lowest order
ρ, $\rho\theta$	$H_2 \otimes \mathrm{DG}_0$	Q_2 Radau × constant per layer
w	$H_2 \otimes \mathrm{CG}_1$	Q_2 Radau × linear, values at the interfaces
$\mathbf{u}$	$U_1 \otimes \mathrm{DG}_0$	RT_1 × constant per layer

Together $(\mathbf{u}, w)$ lies in a space whose divergence maps onto that of ρ, so the discrete continuity equation is exact per kite column. θ lives in the layers with ρ (Lorenz staggering, as in ICON). Every height node carries one column of ρ, $\rho\theta$ and w; for the Radau nodes these columns are both degrees of freedom and quadrature points, so the vertical problem decouples into independent columns. The mass matrices are the tensor products $M_h \otimes \mathrm{diag}(\Delta z)$ (diagonal), $M_u \otimes \mathrm{diag}(\Delta z)$ (block diagonal) and, for w, M_h times the consistent tridiagonal CG_1 mass, whose bandwidth is that of the coupling through ρ and $\rho\theta$.

7.3 Horizontal terms: the vector-invariant form

Per layer the horizontal terms are those of the shallow-water scheme (3) with ρ in place of h:

$$\begin{aligned} \mathbf{F} = \Pi_U(\rho\mathbf{u}), \qquad M_u\dot{\mathbf{u}} &= -\tfrac{1}{a}B^T K - \Big(\int \tfrac{\zeta+f}{\rho}\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{v}_j\Big)_j - c_p\, M_u\, \Pi_U(\theta\, M_u^{-1} B^T \pi/a), \\ M_h\dot{\rho} = B\mathbf{F}/a, \qquad M_h(\dot{\rho\theta}) &= B\,\Pi_U(\theta\mathbf{F})/a, \end{aligned} \tag{6}$$

with the radius a. With $E = \sum_k \Delta z_k \big[\int \rho|\mathbf{u}|^2/2 + \sum_i M_{h,ii}(c_v\rho\theta\,\pi + \rho g z_k)_i\big]$ one has $\partial E/\partial \mathbf{u} = M_u\mathbf{F}$, $\partial E/\partial\rho = M_h(K+gz)$ and $\partial E/\partial(\rho\theta) = M_h c_p\pi$, and the horizontal terms exchange energy without loss, since the pressure pair cancels through the symmetric $\Theta = (\int \theta\,\mathbf{v}_i\cdot\mathbf{v}_j)$. The vertical coupling $-\boldsymbol{\eta}w$ with the horizontal vorticity $\boldsymbol{\eta} = \partial_z\mathbf{u} - \nabla w$ and its counterpart $\boldsymbol{\eta}\cdot\mathbf{u}$ in the w equation are discretized as exact adjoints.

7.4 Vertical terms and time stepping

The vertical terms are those of the C-grid dycores of `dmplex.jl`, one column per height node:

$$\dot{w}_j = -c_p\,\theta_j\,\frac{\pi_j - \pi_{j-1}}{\Delta z^h_j} - g - r_j w_j, \qquad \dot{\rho}_k = -\frac{M_{k+1} - M_k}{\Delta z_k}, \qquad (\dot{\rho\theta})_k = -\frac{\tilde{\theta}_{k+1}M_{k+1} - \tilde{\theta}_k M_k}{\Delta z_k},$$

with $M_j = \rho_j w_j$ and $\tilde{\theta}_j$ the third-order upwind interface value after a Crank–Nicolson vertical advection of θ, as in ICON-IAP. The time step is the Wicker–Skamarock RK3 scheme [10] with HEVI stages: explicit horizontal momentum, a forward–backward update of ρ and $\rho\theta$ with the new velocity, and an implicit vertical solve per column, off-centred with $\beta = 0.65$ and linearized around the Exner pressure of the step, which reduces to one tridiagonal system in w. The initial states are in the discrete hydrostatic balance of these operators, so an atmosphere at rest stays at rest to round-off.

8 Stability of the horizontal transport

The 3D model with (6) conserves energy and keeps the rest state, but is unstable in a flow. Three different causes had to be found. The tool is the spectrum of the linearized horizontal tendency of one layer; $\max \mathrm{Re}\,\lambda > 0$ is a growing mode of the semi-discrete equations, independent of the time step.

8.1 Quadrature of the velocity mass

The velocity mass was first assembled with an exact rule (degree 6), the projections $\Pi_U(h\mathbf{v})$ with the 3×3 rule of the nonlinear terms. On non-affine kites the metric $J^TJ/\det J$ is rational, the two differ, and $\Pi_U(c\,\mathbf{v}) \neq c\,\mathbf{v}$ for a constant c ($1.4\cdot 10^{-2}$ on hexagon kites). Then the $\rho\theta$ flux $\Pi_U(\bar{\theta}\mathbf{F})$ is not $\bar{\theta}$ times the mass flux: the horizontal fluxes create sub-cell entropy at rest. The energy is still conserved, but the Casimir $\int \rho f(\theta)$ that makes the stratified rest state stable is not, and a sub-cell mode lives on the released potential energy. On the affine kites of the cubed sphere both rules are exact and the mode is absent. Assembling M_u with the rule of the projections restores $\Pi_U(c\,\mathbf{v}) = c\,\mathbf{v}$ to round-off and removes the mode.

8.2 Projected transport of θ

A passive tracer ρq transported like $\rho\theta$, $M_h(\dot{\rho q}) = B\,\Pi_U(q\mathbf{F})/a$, grows exactly like θ' in a one-layer flow, although its equation is linear and has no feedback: the transport itself is unstable. The spectrum of the transport of a height field with a prescribed velocity (Williamson case 2, $\mathrm{RT}_1 \times Q_2$), in units of $u_0/\Delta x$:

$\max \mathrm{Re}\,\lambda$	hexagons	cubed sphere	triangles
projected flux $B\,\Pi_U(\rho\mathbf{u})$	0.700	0.278	1.473
pointwise flux, interior trace	4.757	1.245	2.428
DG, central trace	0.001	0.000	0.002
DG, upwind trace	0.000	0.000	0.000

The projected flux tests $\rho\mathbf{u}$ with the discrete gradient $M_u^{-1}B^T\eta$ of the test function, which carries the jumps of η across primal edges spread over the dual-cell mass blocks, so $\langle\dot\rho,\eta\rangle+\langle\dot\eta,\rho\rangle\neq\int\mathbf{u}\cdot\nabla(\rho\eta)=0$: a consistency defect at the grid scale, independent of the quadrature (degree 10 changes nothing) and largest on the irregular kites of triangles. In shallow water and in the 3D model the gravity-wave coupling to the momentum keeps ρ bounded; a transported quantity without that feedback (q, θ') sees the bare operator. The remedy is the DG transport with the kite mass flux,

$$\langle\dot{(\rho q)},\eta\rangle=\frac{1}{a}\Big(\sum_K\int_K q\,\mathbf{F}\cdot\nabla\eta-\sum_{F\in\mathcal{F}_p}\int_F\hat q\,(\mathbf{F}\cdot\mathbf{n}_K)\,[\![\eta]\!]\Big),\tag{7}$$

over the primal faces $\mathcal{F}_p$ with the central or upwind trace $\hat q$. For $q=1$ it is $B\mathbf{F}$, so $\rho q=\rho$ is kept; it is skew in q (central) or dissipative (upwind). The pressure gradient uses its adjoint, so the energy exchange stays exact, and the horizontal advection of w is DG in advective form for the same reason. In the one-layer thermal shallow-water test (Williamson case 2, random $\theta'=10^{-8}$) θ' now stays at 10^{-8} on all three meshes, instead of overflowing within a day on hexagons and triangles.

8.3 Supercritical flow

With the DG transport of θ the steady thermal wind still grows in 3D, on all three meshes and faster on triangles. Ablations show that the growth needs neither shear nor vertical advection nor rotation, but more than one layer: one layer in solid-body rotation stays at rest, five layers grow. The internal gravity waves of vertical mode m in an atmosphere of depth H and buoyancy frequency N travel at about $c_m=NH/(m\pi)\approx 105/m\,\mathrm{m/s}$ for $H=30\,\mathrm{km}$, so the higher modes are slower than the jet. This is reproduced by one layer with a uniform flow $(1,0.3)$ on the periodic unit square and the shallow-water equation of state with wave speed c: it is stable for $c>|\mathbf{u}|$ and grows for $c<|\mathbf{u}|$, at a rate roughly proportional to c. At $c=0.5$:

$\max \mathrm{Re}\,\lambda\,\Delta x/\lvert\mathbf{u}\rvert$	full	momentum (ρ frozen)	continuity ($\mathbf{u}$ frozen)
quadrilaterals	3.49	0.77	0.77
triangles	6.08	5.48	5.49

Both transports of (6) are unstable on their own, the vector-invariant momentum transport (PV flux and ∇K) and the projected continuity $B\,\Pi_U(\rho\mathbf{u})$. In subcritical flow the fast gravity-wave coupling and the conserved energy keep the coupled system neutral; this is why all 2D tests and the one-layer 3D tests pass. In supercritical flow the perturbations can draw energy from the mean flow. In the energy norm $E=\mathrm{diag}(HM_u,gM_h)$ the symmetric part of EJ is 35 % of its skew part, almost all of it in the momentum block; the continuity block contributes 0.3 %. Both blocks must be skew-symmetric.

DG transport of momentum and density. With `horizontal_advection=:dg`, ρ, $\rho\theta$ and the tracers are transported by (7) with the pointwise velocity $\mathbf{F}=\mathbf{u}$ and $\hat q$ the density trace, and the momentum equation is solved in advective form with the density-weighted velocity mass:

$$\int\rho\,\dot{\mathbf{u}}\cdot\mathbf{v}=-\frac{1}{a}N(\mathbf{u};\mathbf{v})-\int\rho f\,(\hat{\mathbf{r}}\times\mathbf{u})\cdot\mathbf{v}-\frac{c_p}{a}\Big(\sum_K\int_K\rho\theta\,\mathbf{v}\cdot\nabla\pi-\sum_{F\in\mathcal{F}_p}\int_F\widehat{\rho\theta}\,(\mathbf{v}\cdot\mathbf{n}_K)[\![\pi]\!]\Big)$$
$$-\sum_{j=k,k+1}\frac{\Delta z_j^h}{2\Delta z_k}\int\rho\,w_j\,\partial_z\mathbf{u}_j\cdot\mathbf{v},\tag{8}$$

$$N(\mathbf{u};\mathbf{v})=\sum_K\int_K\rho\,\mathbf{v}\cdot(\mathbf{u}\cdot\nabla)\mathbf{u}-\sum_{F\in\mathcal{F}}\int_F\hat\rho\,\{\{\mathbf{u}\cdot\mathbf{n}\}\}\,[\![\mathbf{u}]\!]\cdot\{\{\mathbf{v}\}\}+\alpha\sum_{F\in\mathcal{F}}\int_F\hat\rho\,\frac{|\mathbf{u}\cdot\mathbf{n}|}{2}\,[\![\mathbf{u}]\!]\cdot[\![\mathbf{v}]\!],$$

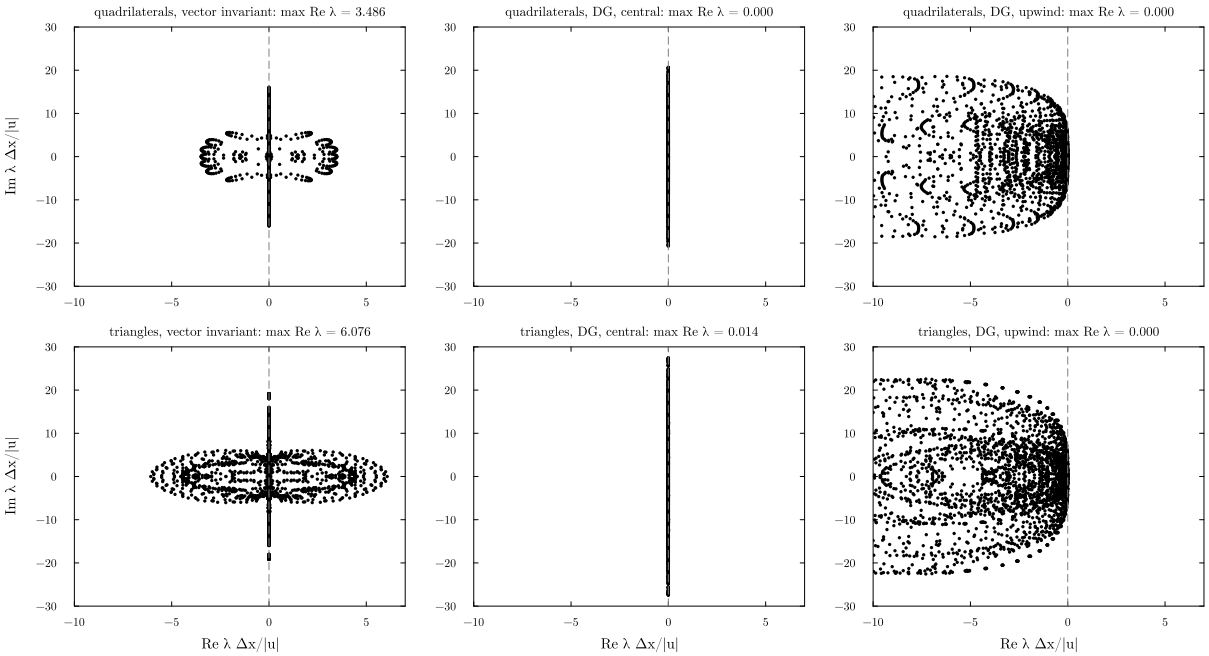


Figure 4: Spectra of the linearized one-layer tendency in a uniform supercritical flow ($|\mathbf{u}| = 1.04$, wave speed $c = 0.5$, periodic 6×6 mesh), in units of $|\mathbf{u}|/\Delta x$: vector-invariant form (6) and DG form (7)–(8) with central and upwind traces, on quadrilaterals (top) and triangles (bottom).

over all four faces $\mathcal{F}$ of every kite (across primal faces only the tangential component jumps, across dual faces the whole velocity), $\hat{\rho}$ the mean of the two traces and $\alpha = 1$ for upwinding. The pressure term is the adjoint of the DG $\rho\theta$ transport, so the exchange with the internal energy stays exact. The volume term is metric free except for the derivative of the Piola factor, $(\mathbf{u} \cdot \nabla)\mathbf{u}\, dA = \sum_k \hat{u}_k\, \partial_{\xi_k}(P\hat{\mathbf{u}})\, d\xi$ with $P = J/\det J$, $\partial_{\xi_k}(P\hat{\mathbf{u}}) = (\partial_{\xi_k} P)\hat{\mathbf{u}} + P\partial_{\xi_k}\hat{\mathbf{u}}$, and $\partial_{\xi_k} P$ from a Richardson extrapolation of central differences of the cell map. The density-weighted mass $\int \rho\, \mathbf{v}_a \cdot \mathbf{v}_b$ is assembled with the rule of the nonlinear terms and factorized per dual cell (Cholesky), so it stays local. For uniform ρ and divergence-free $\mathbf{u}$ the central part of N is skew; the upwind penalty only removes energy. Since the kinetic energy of the advective form is not exactly the K of the pressure term (the projection of K into H_2), the scheme no longer conserves energy exactly; the upwind term supplies the dissipation.

Figure 4 shows the spectra of the one-layer test at $c = 0.5$. The vector-invariant form has growing modes up to $3.49\,|\mathbf{u}|/\Delta x$ on quadrilaterals and $6.08\,|\mathbf{u}|/\Delta x$ on triangles. The DG form with the central trace is neutral on quadrilaterals and nearly so on triangles (0.014, from the quadrature on non-affine kites); with the upwind trace $\max \mathrm{Re}\,\lambda = 0$ on both. The DG advection has the explicit limit of DG with RK3, about 0.1 in the advective Courant number per kite; the 3D runs below are at about 0.02. It costs about 20 % more per step than (6).

9 3D results

All runs use $\mathrm{RT}_1 \times Q_2$ horizontally, 30 equidistant layers up to 30 km, Rayleigh damping of w above 22 km, the DG transport with upwind traces for θ, ρ, $\mathbf{u}$ and w, and no explicit diffusion. They are serial CPU runs.

9.1 Thermal wind

A steady rotating wind in exact gradient-wind balance, $1/T = (1 + \varepsilon \sin^2\varphi\, \mathrm{sech}^2(z/z_s))/T_0$ with $T_0 = 300\,\mathrm{K}$, $\varepsilon = 0.1$, $z_s = 12\,\mathrm{km}$ and the same surface pressure everywhere; the westerly reaches about 24 m/s. A good model keeps it at rest in the rotating frame. Figure 5 compares the kite

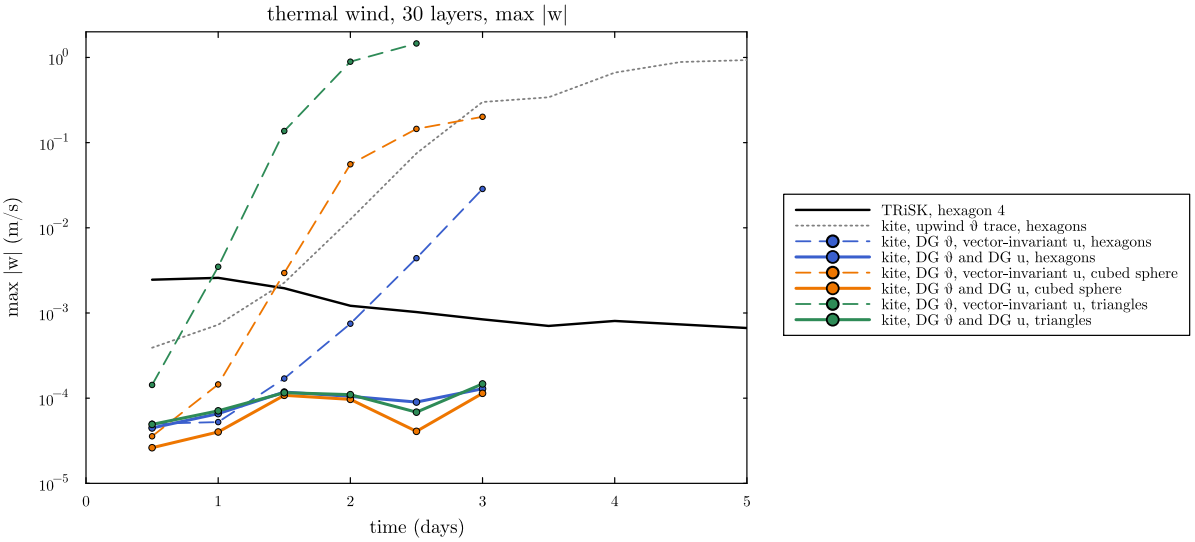


Figure 5: Thermal wind: max $|w|$ of TRiSK (hexagon 4, 40 962 cells) and of the kite model on hexagons 3 (23 682 columns), the cubed sphere 12 (21 600) and triangles 3 (24 320), with the upwind θ trace (grey), the DG θ transport with the vector-invariant momentum (dashed) and the DG transport of θ, ρ and $\mathbf{u}$ (solid).

variants with TRiSK on a finer mesh. With the vector-invariant momentum, max $|w|$ grows on all meshes, to $3 \cdot 10^{-2}$ (hexagons), 0.20 (cubed sphere) and NaN (triangles, day 2.5). With the DG momentum it stays at about 10^{-4} m/s on all three meshes, max $|\theta'|$ at $2 \cdot 10^{-2}$ K and max $|\Delta p_s|$ at $5 \cdot 10^{-2}$ hPa, an order of magnitude below TRiSK on hexagon 4.

9.2 Baroclinic wave in an f-plane channel

The channel of Ullrich, Reed and Jablonowski [8] is 40 000 km long and 6000 km wide, with a jet of 35 m/s, $f_0 = 1.0313 \cdot 10^{-4}\,\mathrm{s}^{-1}$, flat ground and a Gaussian perturbation of u (1 m/s, 600 km). The kite model has no walls, so we use a doubly periodic box of twice the width with the mirrored channel and the easterly jet; $v = 0$ at the walls then holds by symmetry, and the unperturbed state is an exact steady state. For this setup linear quasi-geostrophic theory gives the most unstable wave at zonal wavenumber 10 (4000 km) with the growth rate $6.59 \cdot 10^{-6}\,\mathrm{s}^{-1}$ [8]. Figure 6 shows the amplitude of this wave in the surface pressure on 60×18 cells (667 km), dt = 200 s. After an adjustment of two days it grows at 5.9–$6.6 \cdot 10^{-6}\,\mathrm{s}^{-1}$, reaching $6.58 \cdot 10^{-6}$ at day 11, and saturates after day 12. The triangles follow the quadrilaterals to plotting accuracy; with the vector-invariant momentum the triangle run is dominated by a grid-scale mode from day 1. Figure 7 shows the wave in the surface pressure and the temperature at 500 m. It grows as a packet downstream of the perturbation, and from day 12 the warm air is drawn into tongues that roll up by day 14 with a minimum surface pressure of 955.6 hPa. On the triangles the minima are within 0.7 hPa (979.4 hPa at day 12, 955.8 hPa at day 14). From day 14 they show small ripples in the temperature at the warm edge of the front.

9.3 Baroclinic wave on the sphere

The baroclinic wave of Ullrich et al. [7] with the perturbation of DCMIP 2016, on hexagons 3, the cubed sphere 12 and triangles 3 (about 23 000 columns, roughly 130 km between height nodes), dt = 150 s. Minimum surface pressure (hPa):

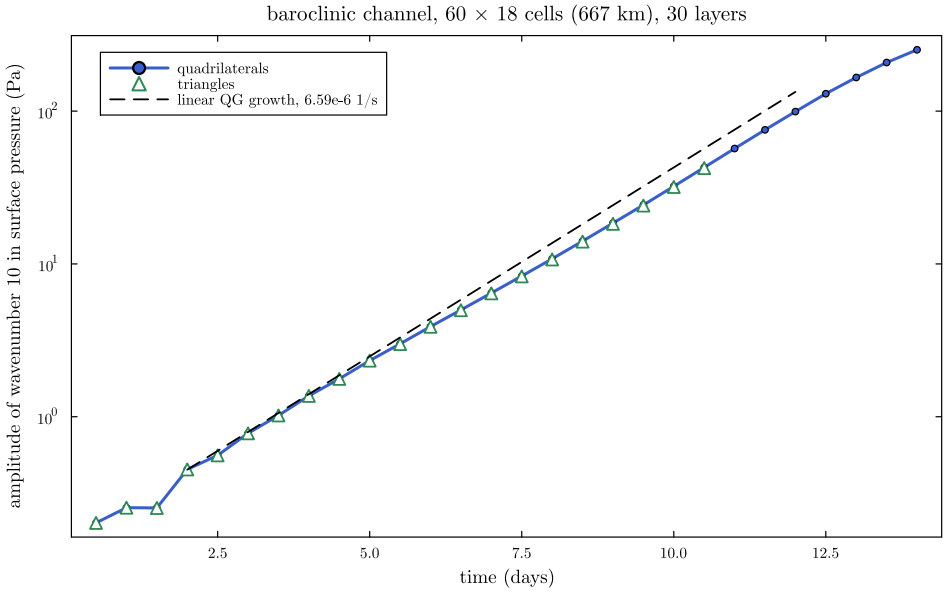


Figure 6: Baroclinic channel: amplitude of zonal wavenumber 10 of the surface pressure on 60 × 18 quadrilaterals (27 000 columns) and triangles (41 040 columns), and the growth rate of linear QG theory.

day	6	7	8	9	10
kite, hexagons (23 682 columns)	993.5	984.3	968.3	941.6	925.8
kite, cubed sphere (21 600)	993.4	984.1	966.9	942.7	926.3
kite, triangles (24 320)	993.4	984.7	967.3	936.0	925.4
TRiSK, hexagon 6 (40 962 cells)	991.7	982.3	963.1	940.4	925.1

The three meshes agree to 1 hPa at day 10 and with TRiSK at a similar resolution (Figures 8 and 9); mass is conserved to 10^{-15}. The maps show the Q_2 fields evaluated inside the kites, not the nodal values. The earlier lowest-order kite run (Q_1 heights) on hexagon 3 reached only 976.6 hPa, with max $|w|$ near 1 m/s from grid-scale noise.

10 Conclusions

The kite spaces give compatible finite elements on arbitrary polygonal meshes with diagonal height and block-diagonal velocity mass matrices. In shallow water the scheme conserves mass and energy exactly, converges at second order on hexagonal, triangular and quadrilateral meshes and runs on GPUs and with MPI. The direct product with the layers of a vertical grid keeps the column structure of HEVI schemes, but its horizontal transport must be built with care: the velocity mass must use the rule of the projections, scalars must be transported by DG with the kite mass flux, and in supercritical flow, which the higher vertical modes of a stratified atmosphere always are, the energy-conserving vector-invariant momentum transport is unstable. A DG transport of momentum and density with the pointwise velocity and upwind traces removes this instability on all meshes. With it the model is quiet on the thermal wind, reproduces the linear baroclinic growth rate in a channel and gives the same baroclinic wave on three different meshes.

Open points: the DG momentum form is not energy conserving, because the kinetic energy of the advective form differs from the projected K of the pressure term; whether an energy-conserving skew-symmetric form with the same stability exists is open. The DG terms are serial CPU loops so far, so the 3D model has not been run at high resolution or on GPUs. Higher vertical order and the reconstruction of θ inside the layers are still to do.

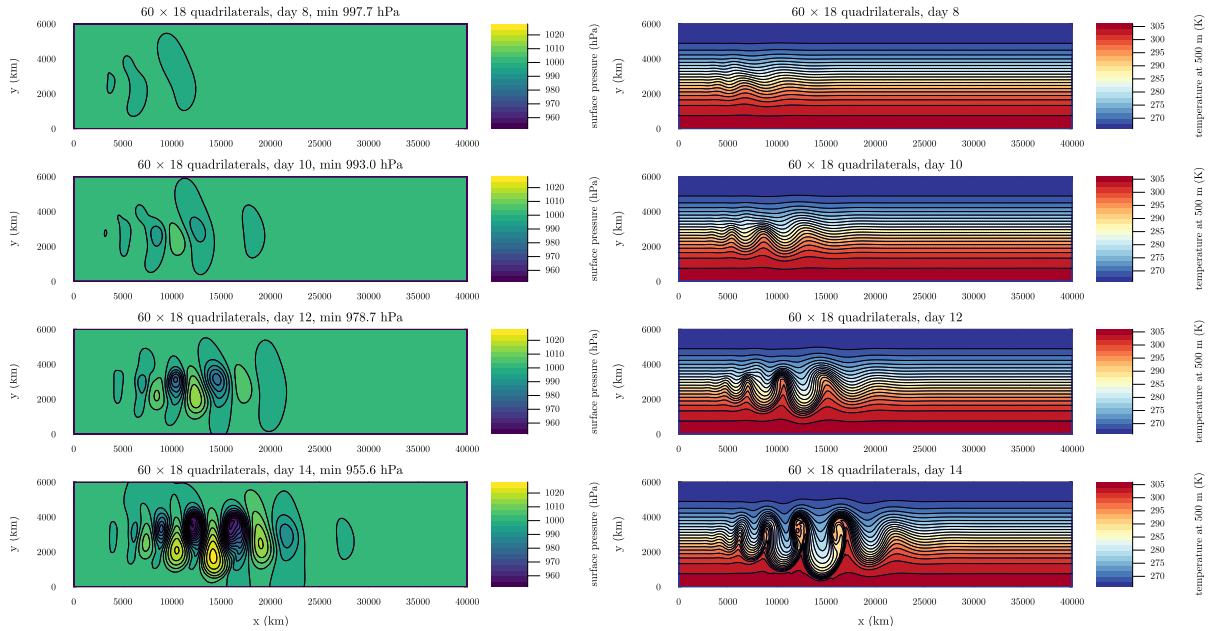


Figure 7: Baroclinic channel on 60×18 quadrilaterals: surface pressure (left) and temperature at 500 m (right) at days 8, 10, 12 and 14, the channel half $0 \leq y \leq 6000$ km of the doubly periodic box.

Code. `dmplex.jl` (`https://github.com/OsKnoth/dmplex`), the models `NonlinearKiteShallowWater` and `KiteNonhydrostatic`. The examples `galewsky.jl`, `bickley.jl`, `thermal_wind.jl` and `baroclinic_channel.jl` run the test cases; the figure scripts are in `docs/paper/scripts`.

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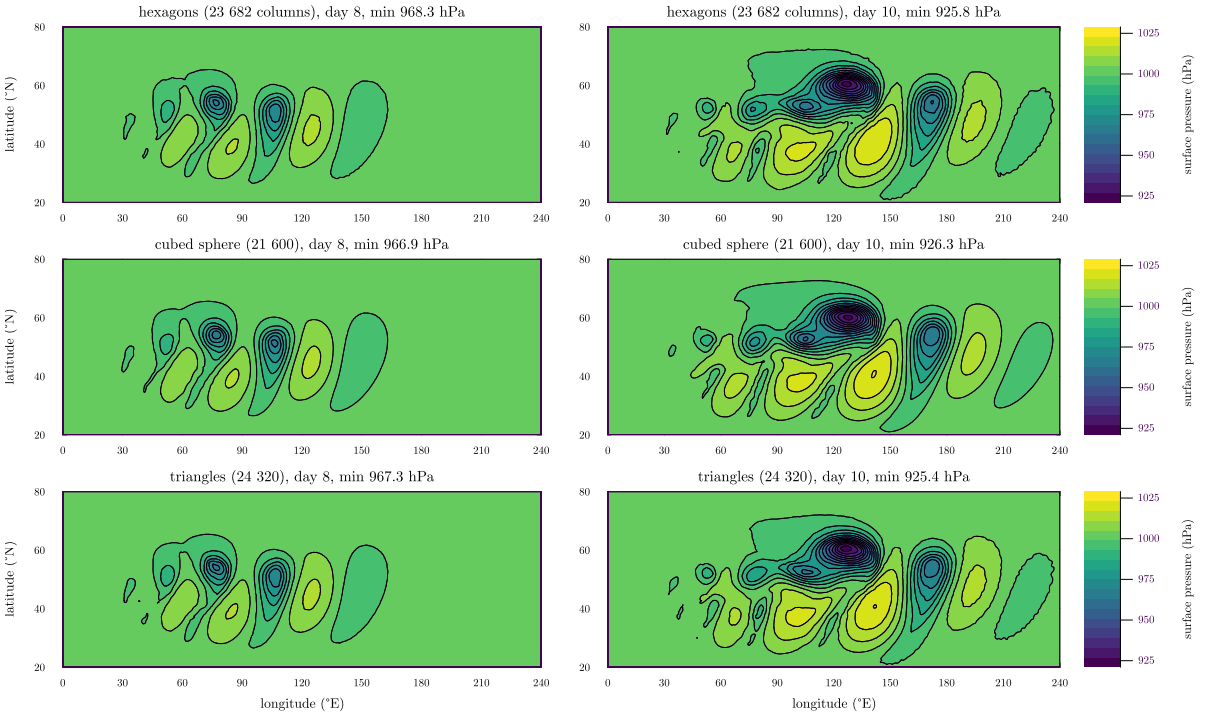


Figure 8: Baroclinic wave, surface pressure at days 8 and 10 on the three kite meshes (contour interval 6 hPa).

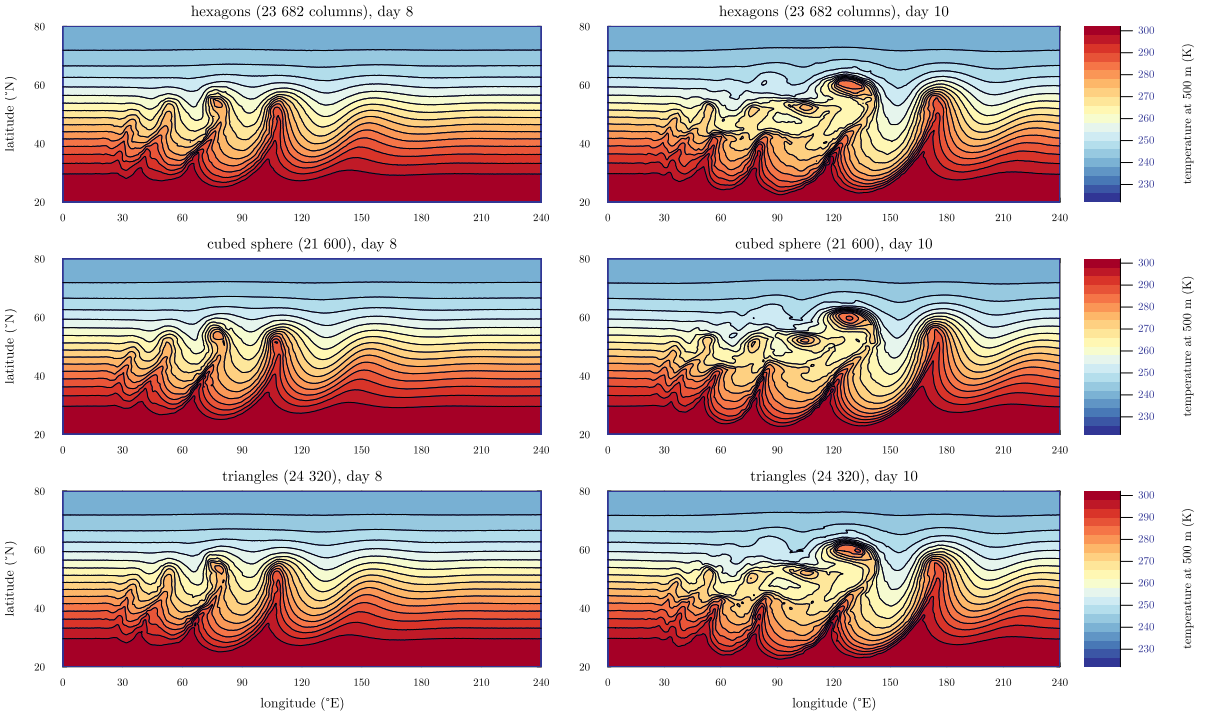


Figure 9: Baroclinic wave, temperature of the lowest layer (500 m) at days 8 and 10 on the three kite meshes (contour interval 4 K).

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